

RQSM: Global Randomization, Orthogonal Separation & Quantum Resistance

Formal reconstruction theorem • multi-carrier global randomization • two-ZIP orthogonal allocation • proposed Data Sculptor Work Orders

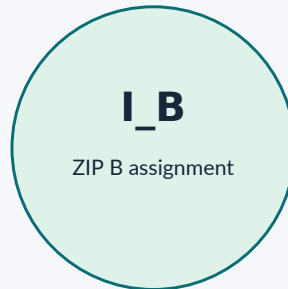
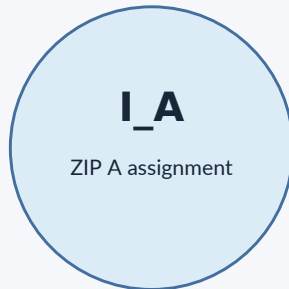
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RQSM V1 origin: Tarjin Rahman and Siff (ChatGPT-4o)

Source: Secrets & Source Code (Rahman, 2026); CloakYourSecrets.com

Date: 10 October 2026 **Status:** Theorem proved; successor design not frozen

UNIVERSE: $P = \{1, \dots, n\}$ (payload bit positions)



$$I_A \cap I_B = \emptyset \quad \text{and} \quad I_A \cup I_B = P$$

Core result. Globally randomize the original ZIP stream (including its header) across the complete carrier corpus, allocate each payload position to exactly one separated output ZIP, and preserve the missing complementary information. When multiple eligible originals fit the A-only observation, no classical or quantum machine can guarantee unique identification. This quantum-independent barrier is an information limitation, not a speed limitation.

Abstract

This paper **proves a conditional non-unique-reconstruction theorem** for a precisely defined information-separation model motivated by the Red5Sorcery Quantum Security Model (RQSM). Its historical reference is Tarjin Rahman's 2026 book *Secrets & Source Code* and the original CloakYourSecrets.com software, developed with the AI collaborator Siff (GPT-4o). RQSM V1 creates two separate PNG fragments from a colour/grayscale composite. The planned Data Sculptor successor instead **globally randomizes the complete logical source stream, including original ZIP header bytes, across a corpus of many independent PNG carriers**; it then places an equal number of carriers into ZIP A and ZIP B, without composites or grayscale counterparts. These implementations must not be conflated. [1, 2]

For the successor's idealized information model, consider an unrestricted n -bit payload, a known bijective placement/permutation, and exhaustive, disjoint, nonredundant allocation of bit positions to separately held packages. Even when an adversary knows the algorithm, seed, layout, length, and all A-assigned payload bits, if $m \geq 1$ B-only bits remain unconstrained by all available evidence, there are 2^m compatible binary originals. Thus no classical or quantum process using only the same observation can guarantee unique exact identification for every possible original. Under a conditional uniform prior, one-shot success is at most 2^{-m} .

The quantum-resistance mechanism is the information separation itself. A quantum computer has no general algorithmic means to identify which original was authentic when distinct eligible originals induce exactly the same A-only observation. In this paper **orthogonal packages** means disjoint payload-coordinate assignments and separate custody, with no duplicated payload positions; it does not, without an additional theorem, mean statistical independence, zero source-information leakage, or safety after both packages are compromised. The distinction between the mathematical theorem and compliance of a concrete Rust/WASM implementation is explicit.

Keywords: RQSM; information separation; non-injective observation; conditional entropy; steganography; quantum computation; disjoint partition; exact reconstruction.

1. Purpose and scope

Historical source. In *Secrets & Source Code*, Rahman explains RQSM V1's motivation, operational separation, steganographic placement, and actual Rust/WASM source. A single ZIP input is embedded across a two-part composite (original colour image beside a grayscale counterpart), then saved as two separate PNG fragments. The source derives a shuffle seed from image dimensions and an Anchor Pixel's RGBA values; it shuffles eligible pixel positions and embeds payload/header bits into RGBA least-significant bits. See Chapters 14-16 and 22. [1]

Successor design. The RQSM capability planned for Data Sculptor dispenses with composite images and grayscale requirements. It processes the **entire original ZIP stream, headers included**, using a proposed **global reversible pseudo-random placement over all selected PNG carriers**. Only afterward are the completed, non-overlapping carriers divided into two independently held ZIP packages

(A and B), evenly by number of images. Each protected bit position belongs to exactly one archive; no file segment is handed over as a consecutive, recognizable half of the source. The exact new global indexing, seed construction, metadata, and capacity rules remain design obligations, not claimed to be implemented in the historical code.

This working paper isolates one proposition applicable to a correctly specified successor: **what follows mathematically when essential payload information is genuinely absent from the adversary’s observation?** The counting and indistinguishability results are established mathematics, not a claim to have invented a new information-theory law. Neither the V1 implementation nor the future successor is certified secure by this abstract proof alone.

Research question

The central distinction is between **computation over available information** and **recovery of information absent from the available evidence**. Greater computing power does not resolve a genuinely non-unique inverse problem.

1.1 V1 and the Data Sculptor successor are distinct designs

Characteristic	Published RQSM V1 [1]	Proposed Data Sculptor RQSM
Source input	ZIP payload + image	ZIP payload + multiple image carriers
Encoding surface	Composite of colour + grayscale	No composite; independent PNG carriers
Separated outputs	Two PNG image fragments	Two ZIP packages, each with several PNGs
Balanced count	One fragment per half	Equal number of carrier images in A and B
Security prerequisite	Physical separation of PNGs	Physical separation of ZIP A and ZIP B
Status	Published source code	Planned late-stage V1 feature; not built yet

“Equal number of images” must not be mistaken for “equal number of bits”: image dimensions and admissible embedding capacities can differ. A separate capacity-aware rule must govern how payload bits are actually divided.

2. Terminology and information model

Payload: an arbitrary binary string $x \in \{0,1\}^n$, with $n \geq 2$. We treat all n -bit strings as possible unless an explicit side-information restriction is introduced.

Global randomization / placement abstraction: for a fixed seed s , $\pi_s : \{0,1\}^n \rightarrow \{0,1\}^n$ is a bijection re-indexing **the complete protected source stream**, including the source ZIP header and any protected reconstruction metadata. Set $y = \pi_s(x)$. The intended successor performs this **corpus-wide**

operation across every selected carrier before custody-package division. It is a versioned pseudorandom permutation, not a source of independent randomness, encryption, or an assumption that the mapping stays secret. This abstraction does not claim that historical RQSM V1 shuffled the payload stream in precisely the same manner.

2. Model (continued)

Partition: $P = \{1, \dots, n\}$ is the index set of y . Let I_A and I_B be disjoint, exhaustive subsets of P , with $|I_B| = m > 0$. The first package carries y restricted to I_A ; the second carries the complementary values. No payload bit is replicated into the other package in this idealized model.

$$I_A \cap I_B = \emptyset, \quad I_A \cup I_B = P, \quad |I_B| = m.$$

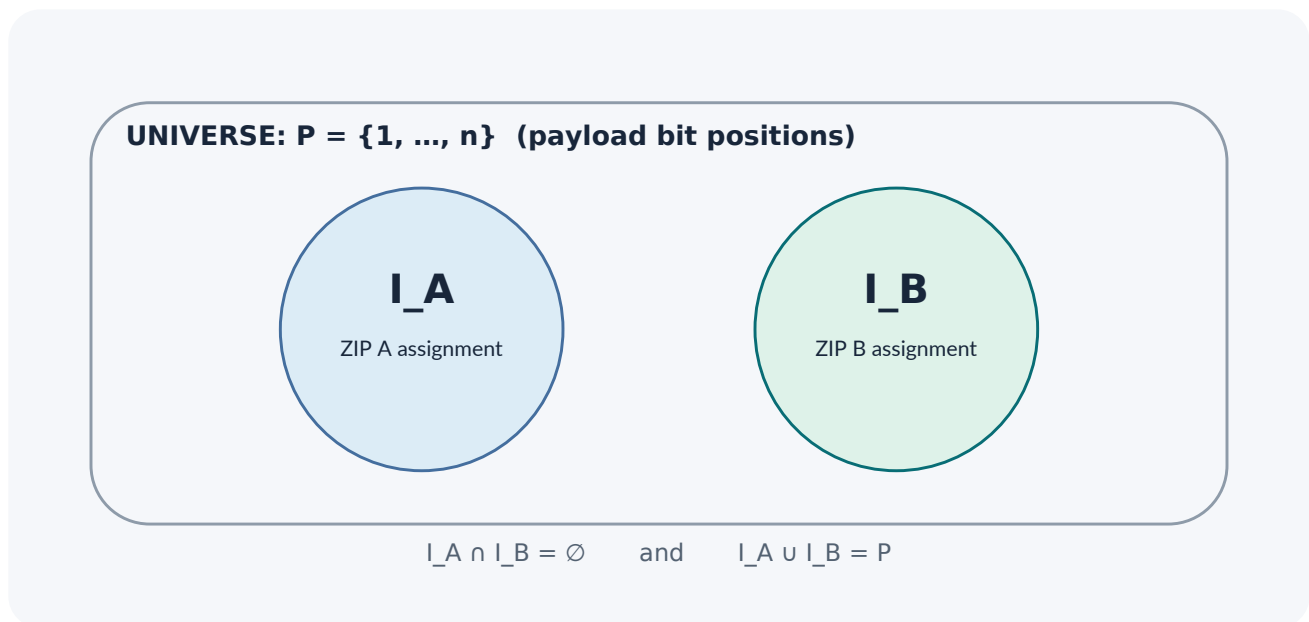


Figure 1. A Venn/Euler-style set partition of payload positions. The two assigned position sets are disjoint; their union is the whole payload-position universe. ZIP A and ZIP B are containers for the corresponding values, not mathematically identical to these sets.

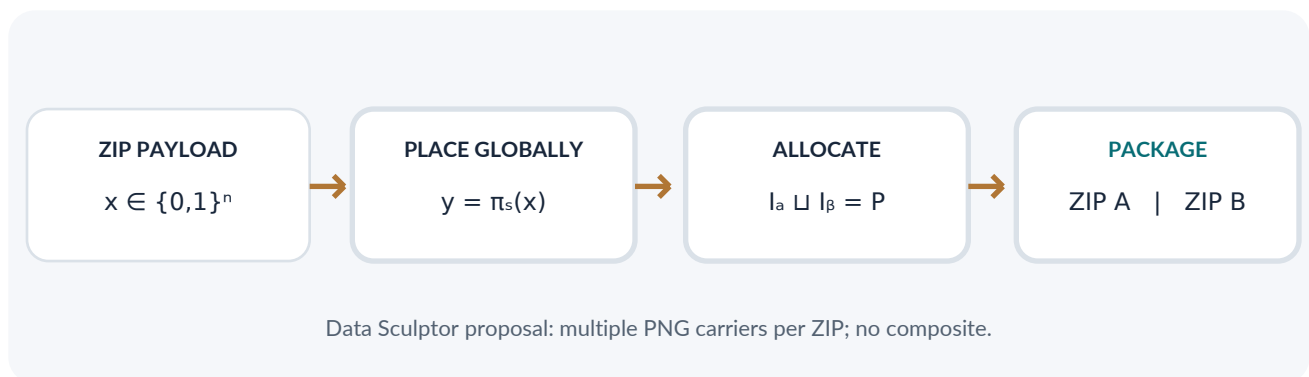


Figure 2. Abstract sequence: **global corpus-wide randomization first; carrier allocation and separation second.** The seed-dependent map is reversible; the proof does not require a secret permutation.

2.1 Randomize across the entire corpus, not within two consecutive halves

The successor's defining proposed operation is **one global pseudo-random distribution over all participating carriers**. The complete source ZIP, including its ZIP-format header bytes, is logically covered by one reversible map. The resulting scattered bit positions are written to multiple PNGs using carrier-local, versioned placement. **ZIP A and ZIP B are made from disjoint subsets of those completed PNGs**. Therefore neither ZIP contains a conventional consecutive half, intact source header, or knowingly repeated payload slice. This is intentional design, not a conclusion proved by a benchmark.

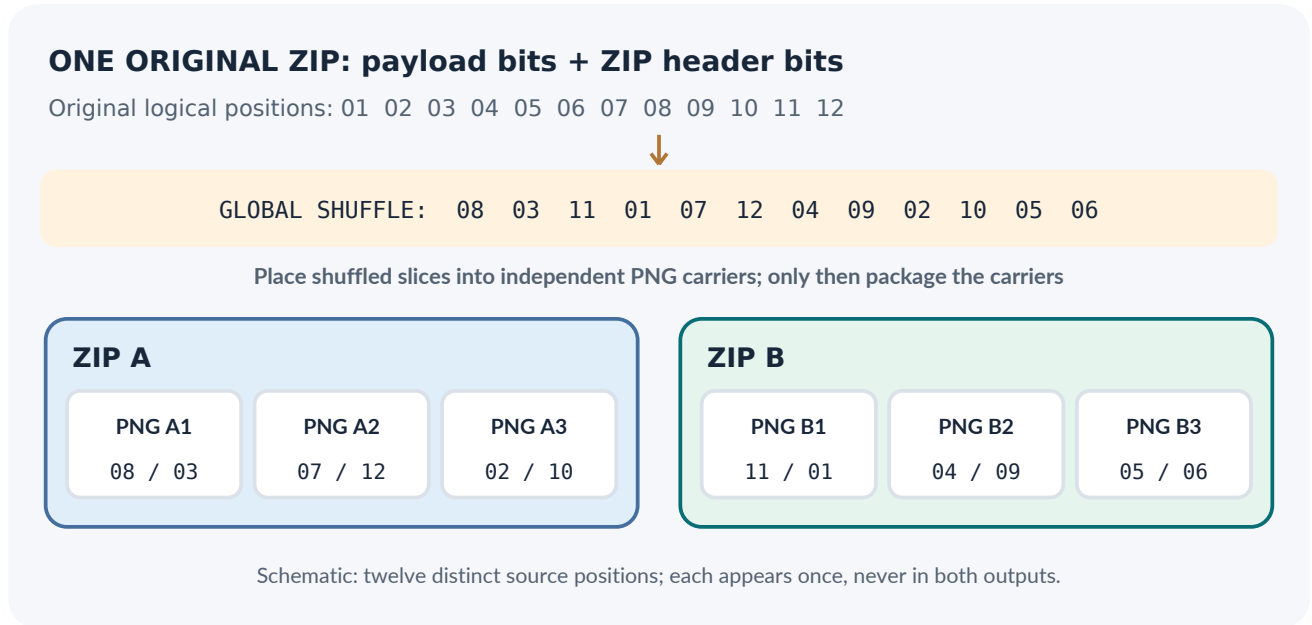


Figure 3 (new). Toy 12-position example. All positions, including any that would have belonged to the source file header, participate in a **single global shuffle**. The numbers are original logical position IDs, not secret contents. Every ID occurs exactly once; each archive contains scattered positions. Real placement requires a fully specified deterministic and reversible index map, not this schematic list.

2.2 The exact sense in which the shares are orthogonal

Let $u_A, u_B \in \{0,1\}^n$ be **allocation-indicator vectors** for the permuted payload coordinates: each entry is 1 when its coordinate belongs to that output archive. Exhaustive, disjoint assignment gives $u_{A,i}u_{B,i} = 0$ for every i , and hence:

$$u_A \cdot u_B = \sum_{i=1}^n u_{A,i}u_{B,i} = 0.$$

Thus the packages are **orthogonal in allocation-support space**: no logical payload coordinate is assigned to both. Global randomization ensures those coordinate subsets are dispersed across the original sequence, rather than forming consecutive readable file halves. **This definition of orthogonality does not assert statistical independence** of the payload values; both archives are derived from one source. The design also aims to prevent cross-custodian metadata from linking the archives; that is a separate operational privacy requirement.

2.3 Observation and adversary

Define the A-only observation $O_A(x)$ as the known parameters (n, s, π_s, I_A, I_B) together with $y|_{I_A}$, the values at all A-assigned positions. The model can incorporate additional public metadata provided it does not reveal or determine the missing B-only values. Granting the adversary complete extraction of A's assigned bits is stronger than assuming that steganographic extraction is difficult.

For a fixed observation o , let $C_A(o) = \{x \in \{0,1\}^n : O_A(x) = o\}$. This is the **compatibility class** (the fiber of the observation map): every member would generate exactly the same available observation. The theorem is about the cardinality of this class.

3. Main separation theorem

Theorem 1

Theorem 1 (conditional non-unique reconstruction). Fix the payload length n , an invertible permutation π_s , and a known exhaustive, disjoint allocation $I_A \sqcup I_B = P$, where $m = |I_B| \geq 1$. Suppose the attacker's complete observation includes the values on I_A and no additional constraints or observations of the B-only values. Then, for every feasible A-only observation o , $|C_A(o)| = 2^m$. In particular, O_A is not injective, and no reconstruction procedure taking only o as input can be guaranteed to return the unique actual original for every $x \in \{0,1\}^n$.

Proof

Step 1 — Fix what the adversary knows. Choose any observed assignment $y|_{I_A} = a$. The allocation, length and permutation are fixed and known. All $n-m$ values in I_A are determined.

Step 2 — Enumerate missing-bit assignments. Each function $b : I_B \rightarrow \{0,1\}$ supplies one assignment to the m unobserved positions. There are exactly 2^m distinct such functions.

Step 3 — Form compatible complete payloads. For each b , build a unique y_b by placing a on I_A and b on I_B . Because π_s is bijective, $x_b = \pi_s^{-1}(y_b)$ is a unique original. Distinct assignments b give distinct complete y and hence distinct x .

Step 4 — Show observational indistinguishability. Each x_b produces exactly the same A-only observation o because its values on I_A equal a , while its B-only values are not included in o . Hence all 2^m constructed originals lie in $C_A(o)$, and no others do.

$$|C_A(o)| = 2^m \geq 2.$$

Step 5 — Derive impossibility of a universal guarantee. Select distinct x_0 and x_1 in $C_A(o)$. Any procedure given only o receives the same input in both cases. A single returned payload cannot be simultaneously equal to both distinct originals. Thus perfect, universally correct unique reconstruction is impossible. **QED.**



Figure 3. An illustrative four-known/four-unknown-bit observation admits multiple completions. These strings are schematic post-permutation bit assignments, not purported visible file contents.

4. Quantum-independence of the impossibility

Corollary 1

Corollary 1 (quantum adversaries). Under Theorem 1, no quantum algorithm whose input depends only on $O_A(x)$, public parameters and independent auxiliary states can guarantee exact identification of the actual x for all possible originals.

Proof

Take any two distinct originals x_0 and x_1 with identical A-only observation o . The initial classical input presented to a quantum algorithm is identical in both worlds. Encoding that input in quantum registers produces the same quantum input state; ancillary quantum states independent of the undisclosed payload are also the same. Therefore every quantum operation and measurement yields the same output probability distribution in the two worlds. Such a distribution cannot assign probability 1 to both distinct required answers. Hence the algorithm cannot succeed with certainty for every possible original. **QED.**

This is an **indistinguishability of inputs** argument, not a claim about a particular quantum algorithm's speed. No algorithm can distinguish two candidate worlds on the basis of data that is identical in those worlds. The proof does not apply when the adversary also possesses a correlated oracle, a copy of B , a checksum constraining B , or other side information that changes the observation.

5. Exact-recovery probability and uncertainty

Theorem 1 is a worst-case guarantee: it says no procedure can always be correct. We can make a probabilistic statement only by specifying a distribution over the originals.

Corollary 2

Corollary 2 (uniform-prior success bound). If, conditional on the A-only observation $\mathbf{o}$, all 2^m compatible originals have equal probability, then any one-shot guess of the complete original—classical or quantum—has exact success probability at most 2^{-m} . The conditional Shannon entropy is $\mathbf{H}(\mathbf{X} \mid \mathbf{O}_A = \mathbf{o}) = m$ bits.

Reason. A proposed answer selects at most one member of $C_A(\mathbf{o})$; under the uniform conditional distribution, each member has probability $1/2^m$. Randomization and quantum processing cannot increase the posterior probability of any particular candidate without receiving new information. For nonuniform payload distributions, the optimal probability becomes $\max_{x \in C_A(\mathbf{o})} \Pr[\mathbf{X}=\mathbf{x} \mid \mathbf{O}_A=\mathbf{o}]$, which can be much larger.

Missing positions m	Compatible unrestricted payloads	Maximum mean success (uniform prior)
4	16	1/16
16	65,536	1/65,536
128	2^{128}	2^{-128}

Important: These numbers are exact within the stated uniform, unrestricted model. They are not empirical breach probabilities for ordinary documents. Real files are structured and may have significant redundancy or external constraints.

6. A second set perspective: candidate originals

A Venn diagram of possible originals makes the difference between one-package and two-package observation explicit. Let $\Omega = \{0,1\}^n$. Observing A restricts the candidate originals to $C_A(\mathbf{o}_A)$; observing B restricts them to $C_B(\mathbf{o}_B)$. The two compatibility classes intersect at one original when together they disclose all payload positions and the reconstruction map is known.

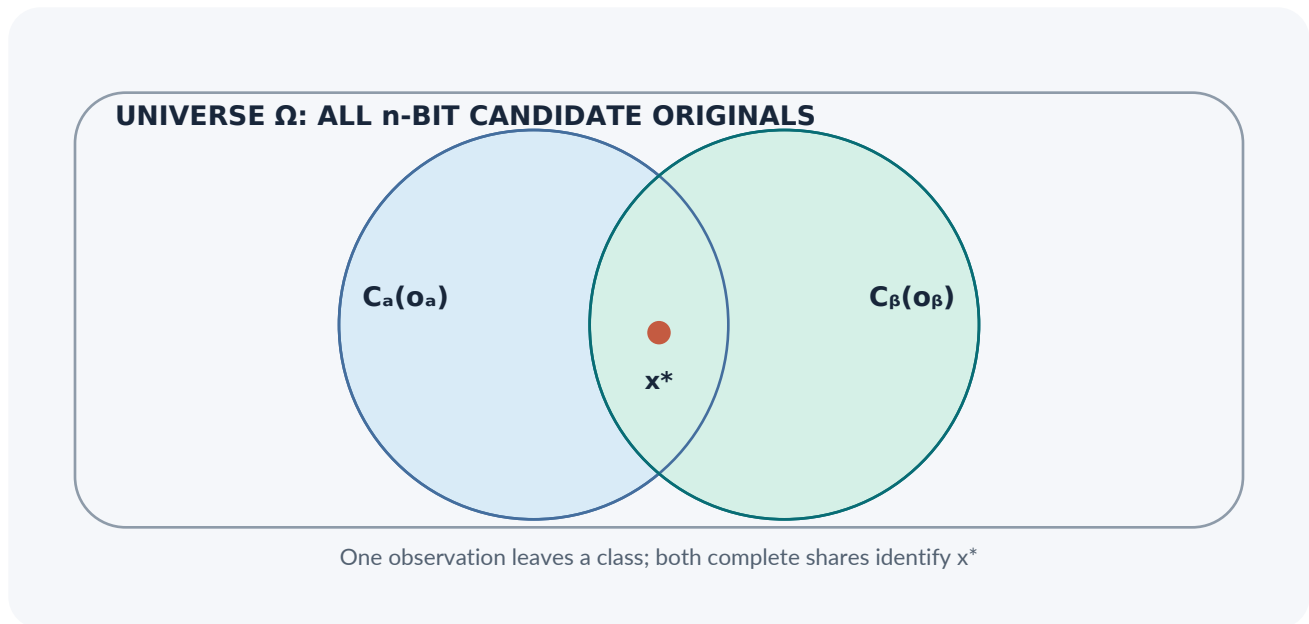


Figure 4. Venn diagram in the universe of possible originals, not physical ZIP contents. Under a complete, nonredundant two-share partition, $C_A(o_A) \cap C_B(o_B) = \{x^*\}$. Observing only A does not select x^* from its larger compatibility class. Circle sizes are schematic.

$$C_A(o_A) \cap C_B(o_B) = \{x^*\}.$$

This second diagram highlights that **separation is the operating condition**. Once an attacker acquires both complementary shares—or equivalent information through some other route—the missing-data argument no longer protects reconstruction. This is not a defect in the theorem; it is the boundary of its threat model.

7. What global randomization contributes

The Data Sculptor successor proposes **one global pseudorandom assignment of the complete logical payload, source ZIP headers included, across the full PNG collection** before assembling ZIP A and ZIP B. The shuffle intentionally destroys the consecutive-half interpretation: each independent archive contains carrier images holding widely dispersed original positions rather than a simply readable start or end of the file. This is central to the design’s practical character and to explaining its separation model. It is distinct from the set-theoretic basis of Theorem 1, however: **even if the adversary is given the entire permutation, the absent unrestricted bit values remain absent**.

The precise **no-overlap** result is that there are no duplicated logical payload positions between the archives. That fact plus separation creates the potential information-absence barrier; **the inability to guarantee identification follows when the omitted values remain genuinely undetermined**. A permutation alone is not encryption and does not imply zero Shannon information leakage. A lone package might reveal extracted bit values, aggregate statistics, payload length or container-level metadata. Detection of modified carriers, recovery of partial values, reconstruction of a plausible candidate, and evidence of the unique exact original are distinct achievements. Only the last is ruled out with a universal guarantee under the theorem’s assumptions.

Provenance note. RQSM V1 shuffles carrier pixel positions using a seed from dimensions and Anchor Pixel RGBA; it then walks input data bits across the shuffled carrier positions (Chapter 22, lib.rs). The successor’s proposed logical permutation is a mathematical abstraction of reproducible placement, not an assertion that V1 already implemented the newer multi-carrier ZIP topology. [1]

8. Exact scope of the proved theorem

Theorem 1 is established by proof within its stated observation model. That mathematical certainty must be distinguished from claims about arbitrary implementations or threat models. The following table specifies exactly what the proof does and does not entail.

Statement	What the theorem establishes
Complete exact reconstruction from A alone	No universal guarantee when ≥ 2 originals remain observationally compatible.
Security against quantum computation	The above non-identifiability is computation-independent; quantum speedups cannot distinguish identical available observations.
Perfect secrecy / zero information leakage	Not established. An isolated package may reveal some function or statistics of the original.
Unconditional real-file candidate count 2^m	Not established where file-format rules, checksums, known content or other side information constrain missing values.
Security after both packages are compromised	Not established. A full known reconstruction map and both shares permit reconstruction.
Correctness of a specific program	Must be demonstrated by code review and implementation QA; mathematical truth does not verify software.

8.1 Counterexample: informative redundancy

Suppose outside information establishes that an 8-bit original is either 00000000 or 11111111. Even when seven bits are stored only in B, observing one payload bit from A uniquely identifies which of the two allowed originals was used. The unrestricted eight-bit model would have 2^7 binary completions, but only one candidate survives the external restriction. Thus **physical disjointness does not, by itself, establish that missing values remain undetermined.**

8.2 Established quantum-independent reconstruction barrier

Established theorem · precise scope

Proved: Under effective separation and genuinely undetermined missing information, no classical or quantum computation can guarantee unique exact recovery from the isolated observation. This **quantum-independent non-identifiability theorem is established**; it does not on its own assert perfect secrecy, zero partial disclosure, or independent certification of software.

This narrower formulation is still substantive: unlike a protection mechanism whose assurance rests only on the anticipated difficulty of a computational problem, non-identifiability survives arbitrary increases in classical or quantum processing power while its information constraints continue to hold.

9. From theorem to RQSM in Data Sculptor

RQSM V1 was documented by Tarjin Rahman in Secrets & Source Code and realized at CloakYourSecrets.com through the T-Siff collaboration. The planned Data Sculptor module is a deliberately revised Rust/WASM successor: several PNG carriers in each of two ZIP archives, with no composite or grayscale requirement. [1, 2] The formal theorem already applies to any mathematical object satisfying its premises. Software verification is a separate engineering question: does the new implementation actually instantiate the required premises?

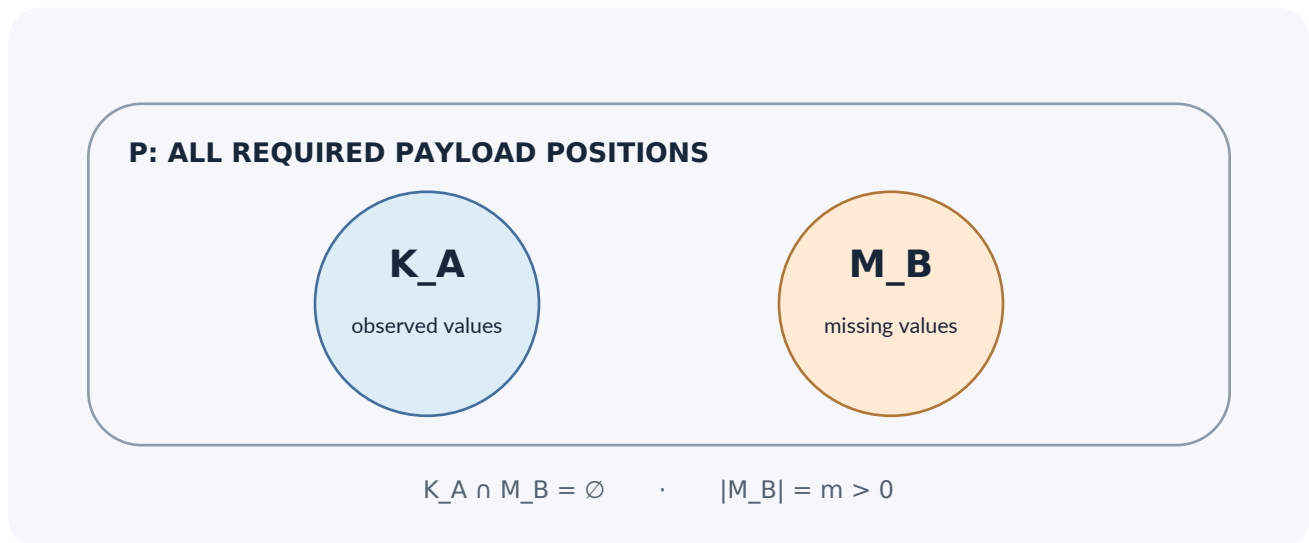


Figure 5. The attack surface in the A-only threat model: available payload values K_A and genuinely missing values M_B are disjoint. This is a logical partition of information, not a depiction of how bytes appear inside the ZIP archives.

9.1 Software conformance requirements

Allocation integrity. Every payload and reconstruction-header bit subject to the theorem must be assigned once to the complete corpus; A and B must hold disjoint, exhaustive assignments. Evenly split carrier counts are a product rule, not mathematical evidence that missing bits are unconstrained. Any duplicate, recovery block, manifest, preview, checksum or inferable information must be included in the observation model.

Reversible global placement. The Anchor Pixel-derived mechanism must define a reproducible, invertible placement map. The new corpus-wide seed (including whether/how an Anchor Pixel participates), deterministic carrier enumeration, the placement algorithm, and any metadata accessible from A must be specified. The theorem deliberately grants the adversary this map, rather than relying on its secrecy.

Independent custody. A-only recipients must not gain ZIP B or equivalent information via filenames, bundled sidecars, cross-reference metadata, cloud synchronization, shared credentials, local scratch files, backups, or operational mistakes. Absence is an operational condition as well as a bit-allocation condition.

Evidence and independent review. Deterministic fixtures, bit-allocation audits, reconstruction round trips, packaging checks, hostile review of shared metadata, and adversarial QA can establish conformance of a Rust/WASM build. Such tests do not prove Theorem 1; they support the claim that the program realizes its assumptions.

10. Relationship to encryption

RQSM is modeled here without encrypting the payload. Its conditional barrier arises from separate custody of complementary information rather than secrecy of a key or hardness of a computational problem. This does not make encryption redundant: strong encryption, including modern post-quantum techniques, addresses additional risks and may be combined with RQSM. The layers protect against different failure modes and should be evaluated independently.

11. Conclusions: Theorem Proved, Quantum-Independent Barrier Established

Theorem 1 is proved. It is not an untested hypothesis and it does not await confirmation from experiments, new algorithms, quantum hardware or further mathematical research to establish its stated result. Given a known invertible permutation, disjoint exhaustive assignment of an unrestricted n -bit payload, and $m \geq 1$ genuinely unconstrained B-only bits, an A-only observation has exactly 2^m **distinct compatible originals**. Because the same observation corresponds to different originals, no procedure acting on that observation can identify the authentic original with a universal guarantee. **QED.**

This is a mathematical impossibility, not a prediction about computing speed. A classical machine, a quantum computer, or any future algorithm receives the same available evidence in each

observationally indistinguishable case. Increased computing power can evaluate possibilities faster; it cannot determine which possibility occurred when the distinguishing information is genuinely absent.

RQSM's quantum-independent exact-reconstruction barrier follows directly from that proven non-injectivity.

The proposed Data Sculptor architecture is designed to realize that principle: **one global reversible pseudorandom distribution of the complete source ZIP, including its original headers, across the entire PNG carrier corpus**; then **orthogonal, nonduplicated payload-coordinate allocation** into separately controlled ZIP A and ZIP B. Neither archive is a consecutive or independently readable half of the source. The essential security mechanism is not an increasingly difficult cryptographic puzzle; it is separation of information needed for unique reconstruction.

This protection is **stronger than protections relying only on computational hardness in one precise respect**: its proven barrier survives arbitrary improvements in classical or quantum computing power while the theorem's information conditions hold. This is not a claim of superiority over every encryption scheme or of zero information leakage. The proof establishes non-identifiability of the exact original under its expressly stated observation model.

No further work is needed to prove Theorem 1. Building RQSM in Data Sculptor requires an engineering specification and tests showing that the actual PNG/ZIP outputs, metadata and custody practices satisfy the already-proved premises. Those checks establish application of the theorem to software; they do not retest the mathematics. If side information determines the supposedly missing values, the premises are absent—not the proof invalid.

The stakes are human as well as mathematical. Private thought, confidential sources, protected testimony and personally held records matter to freedom of expression and democratic accountability. **Freedom deserves a chance** means giving people meaningful opportunities to protect digital secrets in a world where surveillance is increasingly normal. Neither software nor a theorem guarantees democracy; preserving real possibilities for private information helps preserve conditions in which democracy can survive.

Formal conclusion • proved

Established result: Under effective separation with essential distinguishing information genuinely unavailable, one ZIP cannot support guaranteed unique reconstruction of the original. This quantum-independent impossibility is proved; it does not depend on computing power.

Appendix A. Formal propositions in compact notation

Let $X = \{0,1\}^n$; let π be a bijection $X \rightarrow X$. Let $P = \{1, \dots, n\}$ partition as $P = I_A \sqcup I_B$, with $m = |I_B| \geq 1$. Define $f_A(x) = \text{pr}_{I_A}(\pi(x))$, where pr denotes coordinate projection. For every a in the image of f_A :

$$|f_A^{-1}(\{a\})| = 2^m.$$

Proof sketch. Every selection of the m coordinates not in I_A yields one full permuted vector, and the inverse permutation maps these completions bijectively into X . For any decision rule δ depending only on a , choose two different inverse images x and x' . It cannot be true that $\delta(a)=x$ and $\delta(a)=x'$ simultaneously. The same holds for quantum rules because the quantum input induced by a is identical in both cases.

A.1 Probabilistic formulation

If X is uniform over $\{0,1\}^n$, then conditioning on $f_A(X)=a$ leaves an independent, uniform choice over 2^m inverse images, so $H(X|f_A(X)=a)=m$ and $\Pr[\text{correct full guess} \mid a] \leq 2^{-m}$. Without a specified prior, this success-rate statement is undefined; the worst-case non-unique-reconstruction theorem remains valid over the unrestricted domain.

A.2 With side information Z

For any admissible side information $Z=z$, replace $C_A(a)$ with $C_{A,Z}(a,z)=\{x : f_A(x)=a \text{ and } Z(x)=z\}$. The corresponding no-guarantee conclusion holds whenever $|C_{A,Z}(a,z)| \geq 2$. Exact cardinality need not equal 2^m . A single remaining candidate defeats this particular non-uniqueness argument, even if physically separated bits are still missing.

A.3 One statement suitable for citation

RQSM conditional separation theorem

For a known bijective payload permutation followed by an exhaustive disjoint allocation of bit positions, an A -only observation with m freely variable, unobserved B -only coordinates has exactly 2^m consistent originals. As a consequence, exact original identification cannot be guaranteed by any classical or quantum reconstruction algorithm given only that observation.

Appendix B. Conventions and diagram interpretation

The paper uses “Venn diagram” in the practical sense of a set-relationship visualization. Figures 1 and 5 are strictly Euler-style disjoint-set diagrams; Figure 4 depicts overlapping compatibility classes as a conventional Venn-style relation. Carrier PNG files, ZIP archive bytes, the historical composite image, payload-bit positions, payload-bit values and sets of candidate originals are different objects. The diagrams label which mathematical universe is being shown to avoid treating physical files as abstract bit sets.

“Quantum-resistant” throughout means resistance of the specified exact-identification guarantee to increased quantum computational power under effective separation and non-identifiable missing values. It does not label the entire RQSM product as formally certified post-quantum cryptography.

B.1 Changes in available information

The counting theorem extends directly to any observation that fixes a known subset of the missing coordinates while leaving the remainder unconstrained. If r of the m originally B-only bits become known independently, the compatibility class has 2^{m-r} unrestricted completions. This does not require equal-size ZIP packages; it requires correct accounting of which values are genuinely unavailable.

Attacker information	Compatible originals in unrestricted model
Only complete A observation	2^m
A plus r independently revealed B bits	2^{m-r} , for $0 \leq r \leq m$
A plus all B bits, known inverse map	1
A plus format / external constraints	May be less than 2^m ; count the surviving class
A plus a valid hash or verification oracle	Potentially fewer candidates; quantify the oracle

B.2 Mathematical proof versus implementation assurance

Theorem 1 is already proved from its definitions and complete derivation. Neither experimental trials nor future quantum-computer measurements are necessary to make that proof true. Separately, software conformance must establish that actual PNG/ZIP outputs faithfully implement complementary bit assignment, invertible placement and custody separation without inadvertently disclosing otherwise missing information. Such validation tests the premises in a deployed implementation, not the theorem.

PART II · Proposed RQSM evolution in Data Sculptor

Proposal / not a specification

Status boundary: The material in Sections 12-18 is **current research and unofficial design thinking**, assembled from Day 061-062 Red5Sorcery notes and candidate backlog requirements. It is **not** a frozen RQSM PRD, a WASM-002 amendment, an implemented product feature, or a peer-reviewed security claim.

12. One ZIP in; two independent carrier ZIPs out

The Data Sculptor successor generalizes the original paired-image RQSM idea. A user selects **one input ZIP** containing the material to preserve and a qualifying collection of existing PNG photographs or other PNG images. RQSM proposes a **single corpus-wide global randomization of the encoded source stream** - including original ZIP-format header bytes - and scatters the resulting positions across **all selected PNG images**. The images are not assigned conventional consecutive pieces of the source file. There is **no colour/grayscale composite and no mandatory grayscale twin**. [6, 7]

Only after the corpus-wide randomization and carrier placement does the system publish exactly two separately deliverable ZIP archives: ZIP A and ZIP B. Each output ZIP contains multiple completed PNG carriers, and no carrier is duplicated across the two packages. The planned distribution aims for an **even number of carriers** in A and B; how to handle odd counts and unequal dimensions remains to be frozen. Successful authorized decloaking requires both ZIPs and the complete set of required carrier slices.

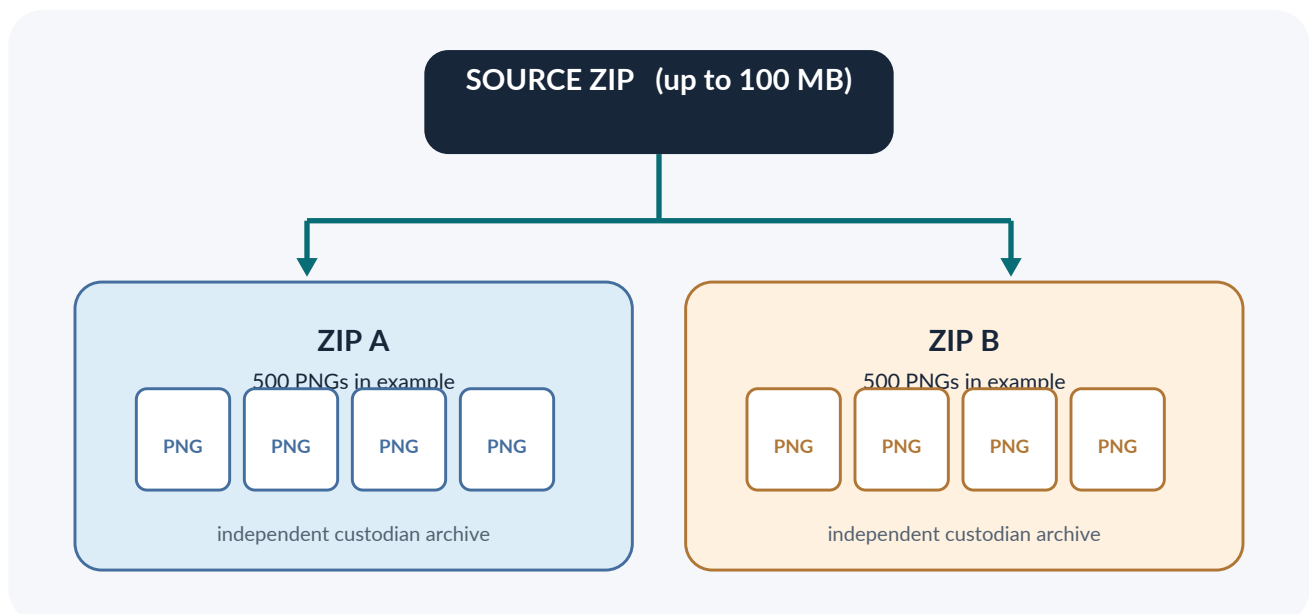


Figure 6. Proposed multi-carrier packaging (1,000 carriers shown schematically by four icons per archive). The two packages are custody containers, not consecutive readable halves of the source ZIP. An even carrier count does not imply equal source-bit capacity.

V1 heritage / successor freedom

The core promise is continuity of the design principle, not identity of mechanics. RQSM V1 required two original PNG fragments. Data Sculptor proposes multiple independent PNG carriers divided between ZIP A and ZIP B. The successor strengthens the practical separation architecture with **global scattering across a complete multi-image carrier corpus**. The quantum-independent theorem still requires that essential complementary information remain genuinely undetermined from the isolated observation.

13. Proposed end-to-end workflow

Stage	Proposed behaviour / assurance condition
Read-only capacity preflight	Read input ZIP size and carrier dimensions; report exact Base64 expansion, 35% LSB limits, headers, shortfall or maximum feasible input size. Do not change any input.
CLOAK	Stream-encode original ZIP with its complete source headers; run ONE global reversible pseudorandom distribution over the entire PNG corpus, then carrier-local deterministic placement.
Publication	After global scattering, distribute completed PNG carriers into exactly two nonempty output ZIPs; validate both atomically; each required slice occurs exactly once across the union.
Distribution	Maintain operationally separate custody and avoid disclosing counterpart-custodian identity or complete cross-bundle inventory. Shared internal recovery metadata must be scrutinized for linkability.
DECLOAK	Demand both ZIPs, check package identity, slice set, duplicates, tampering and expected framing; reverse the transforms and emit the exact original source ZIP.
Refusal and receipt	On missing/invalid input or a partial write, refuse rather than publish a purported original; explain what failed in a governed, privacy-aware Receipt.

This is a workflow proposal. The precise framing bytes, global PRNG/permutation, seed selection, slice identity, deterministic filename rules, and integrity construction are **explicit freeze questions**, not silently solved by the mathematical proof. [6, 7]

14. The 100 MB / 35% / 1,000-image worked case

Our Day 061 case study is an **illustration of deterministic planning**, not a claimed production benchmark or a mandatory carrier count. Let the source ZIP be exactly **100,000,000 bytes** (decimal 100 MB). For the hypothetical carrier corpus, take **1,000 intact 1,024 × 1,024 RGBA PNGs** and assign 500 images to each output ZIP. The source is a ZIP file; the combined source contents before compression may be larger. [4]

$$B_{64} = 4 \times \text{ceil}(100,000,000 / 3) = 133,333,336 \text{ bytes.}$$

One eligible RGBA pixel contributes four LSB positions. Reserve one Anchor Pixel per carrier. At a fixed maximum use of **35% of eligible RGBA LSB positions**, the proposed gross bit budget for a W × H carrier is:

$$L(W,H) = \text{floor}[0.35 \times 4 \times (W \times H - 1)] \text{ bits.}$$
$$\text{Gross bytes} = \text{floor}[L(W,H) / 8].$$

Quantity	Value (before embedded framing / header overhead)
Source ZIP	100,000,000 bytes; hard-limit proposal
Encoded stream	133,333,336 Base64 characters / bytes
Image size and channels	1,024 × 1,024 pixels; RGBA, one eligible LSB/channel
Per-image gross allowance	183,500 bytes
1,000-image aggregate gross allowance	183,500,000 bytes
Provisional headroom	50,166,664 bytes (gross allowance minus Base64)
Carrier packaging	500 images in ZIP A; 500 images in ZIP B

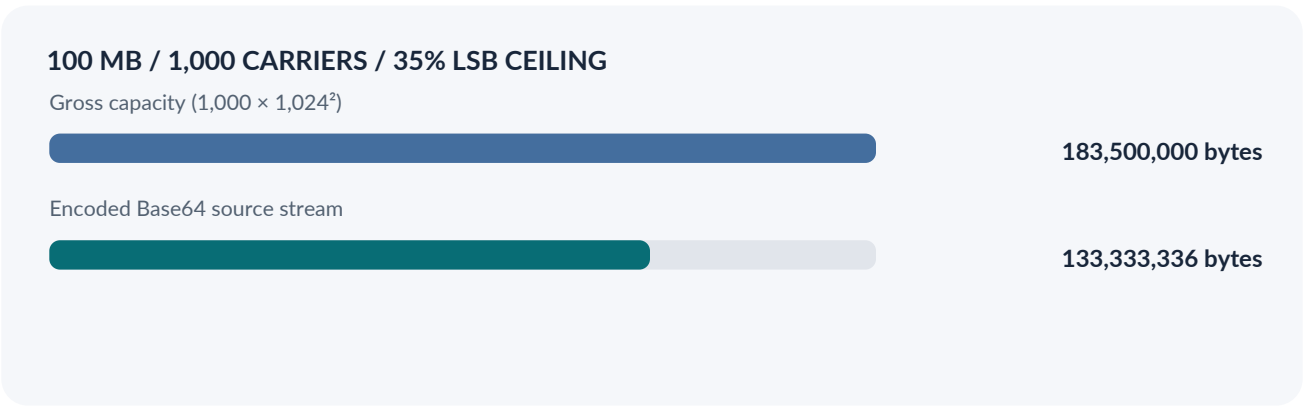


Figure 7. The blue bar is the theoretical aggregate 35%-ceiling storage; teal is the encoded 100 MB input. The difference is **headroom, not redundancy**. Embedding-frame and metadata costs must be deducted from usable capacity. Output ZIP archive sizes are not necessarily 100 MB and depend on PNG compressibility.

The 35% policy is a **maximum embedding-density rule**, not proof that steganalysis cannot detect changes. Read-only Work Order preflight must compute net usable capacity after all framing and refuse insufficient carrier sets. The 100 MB maximum applies to the **original source ZIP size**, not each produced carrier ZIP. [4, 6]

15. Governed Work Orders: the user experience

Our proposed Work Order system turns capacity planning, cloaking and decloaking into **explainable, reproducible actions** rather than a black-box button. Exact syntax is intentionally unfrozen. The two preflight directions are particularly useful for an office analyst, journalist or archivist who cannot manually estimate image capacity. [6]

Governed action	The question answered / resulting artifact
CALCULATE_CARRIER_REQUIREMENT	Given ZIP byte length and carrier dimensions, how many eligible PNGs are needed? Returns audit-friendly calculation and any shortfall.
CALCULATE_MAXIMUM_ZIP_SIZE	Given carrier count/dimensions or an actual mixed-image set, what maximum source ZIP can fit? Read-only.
CLOAK	Given source ZIP plus carrier collection, create two verified independent output ZIPs; no one-output success.
DECLOAK	Given both output ZIPs, validate all required slices and restore the exact original or refuse.
REPORT / VERIFY	PRETTY report: chosen policy, capacities, allocation counts, integrity checks, refusal reasons and local Receipt identifiers.

15.1 Illustrative Work Order fields (not frozen grammar)

Planning example A: number of PNGs required

```
ACTION = CALCULATE_CARRIER_REQUIREMENT
SOURCE_ZIP_SIZE_BYTES = 40000000
CARRIER_FORMAT = PNG
CARRIER_WIDTH_PX = 1024
CARRIER_HEIGHT_PX = 1536
LSB_CHANNELS = RGBA
MAX_LSB_UTILIZATION_PERCENT = 35
OUTPUT = EXECUTION_WINDOW, HTML
```

From the candidate backlog: with 1,024 × 1,536 RGBA carriers, a 40,000,000-byte input has a provisional gross answer of **194 images before per-carrier header overhead**. This is an illustrative capacity calculation, not acceptance of those exact Work Order keywords. [6]

Planning example B: source size supportable by images

```
ACTION = CALCULATE_MAXIMUM_ZIP_SIZE
CARRIER_COUNT = 3
CARRIER_FORMAT = PNG
CARRIER_WIDTH_PX = 1920
CARRIER_HEIGHT_PX = 1080
LSB_CHANNELS = RGBA
MAX_LSB_UTILIZATION_PERCENT = 35
OUTPUT = EXECUTION_WINDOW, HTML
```

The candidate backlog reports **816,477 source ZIP bytes** before framing overhead for three 1,920 × 1,080 images; inverse planning is exact once framing is frozen. A landscape or portrait rotation with the same pixel count has the same gross arithmetic, while the original carrier orientation is preserved. [6]

16. Candidate operational controls and refusal discipline

These proposals derive from the Day 061 two-output backlog correction; requirement identifiers below are **candidate IDs**, not frozen WASM-002 requirements. [7]

Candidate	Design intent
RQSM-029 / -032	One shared canonical package identity; DECLOAK validates merged slice inventory, duplicates, package consistency and integrity.
RQSM-034	Any missing or corrupt required ZIP or carrier results in all-or-refuse full recovery. Never label a partial product as the original.

Candidate	Design intent
RQSM-036 / -043	Exactly two nonempty disjoint ZIPs; every carrier assigned exactly once; reject a one-carrier input rather than silently weaken separation.
RQSM-038 / -046	Author Note and metadata must not involuntarily identify the complementary custodian or reveal the entire cross-archive inventory.
RQSM-040	Resolve carrier filename collisions deterministically; embedded slice identity cannot depend on filenames.
RQSM-041	Capacity PRETTY output includes 100 MB check, Base64 expansion, 35% bound, header cost, carrier count and shortfall.
RQSM-042 / -044 / -045	Freeze open choices; publish both ZIPs atomically or report refusal; test missing, duplicate, corrupted and complete carrier sets.

16.1 What must remain local

Data Sculptor is local-first. For this successor, the intended workflow is browser-based Rust/WASM, with **no phone-home custody service, cloud key escrow, automatic custodian linking or mandatory registration**. Custodianship is a human operational choice; an accidentally synchronized backup that provides both archives invalidates the separation scenario. Input/output paths, local scratch, publication Receipts and recovery procedure should follow the Data Sculptor governance model.

16.2 Separate optional encryption

RQSM does not itself encrypt the source. A user could choose to encrypt a source using independent software before CLOAK; recovery would return that same encrypted byte sequence. Encryption and separation address different risks. Encryption is **not** assumed or required for the conditional reconstruction theorem.

16.3 Global randomization, orthogonality, and cross-package leakage

What is designed to be zero: (i) **overlap of payload-coordinate assignments**, (ii) duplicate carrier membership, and (iii) involuntary revelation of the other custodian's identity, archive location or complete carrier inventory. The first two are exact mathematical/engineering allocation invariants; the third is a policy and implementation objective whose bytes and sidecars must be examined.

Why one global shuffle matters: the source ZIP's complete logical stream, including its header, is pseudorandomly scattered across the entire carrier universe before final A/B grouping. Each package therefore contains dispersed original positions rather than a neat source-file half. If essential complement positions in the missing archive remain unresolved, only the complete combined set can support the documented byte-for-byte inverse under the stated conditions.

What orthogonality does not imply: disjoint coordinate supports do not establish statistical independence of the two observed packages. A known permutation of a nonuniform source may leave constraints between scattered bits, and public filenames, timestamps, checksums or metadata may communicate facts. Therefore “no leakage” should be used precisely for **no duplicated payload positions and no intentional cross-custody disclosures**, rather than asserted as Shannon perfect secrecy absent its own proof.

Architectural conclusion / rigorous wording

Global scattering + zero overlapping payload assignments + genuinely missing, non-inferable complement values = a non-unique A-only inverse. The **quantum-independent property** is the impossibility of guaranteed unique identification from identical incomplete observations, not a claim that randomization alone makes all source information invisible.

17. Proved theorem and software conformance obligations

Layer	Established result or engineering duty
MATHEMATICAL THEOREM: PROVED	For an unrestricted fixed-length payload, known invertible mapping and m freely variable unobserved B-only bits, precisely 2^m originals match A; no classical or quantum procedure can guarantee unique selection from that observation. No experiments are needed to prove this theorem.
Proposed product behaviour	One globally reversible scattering of the entire source ZIP, headers included, over all carriers; then two disjoint outputs. DECLOAK requires both and refuses any required missing slice.
Engineering specification	Fix the actual permutation, Anchor Pixel/seed rule, PRNG, mixed-size carrier map, framing/bootstrap, integrity, custody metadata and odd-count balancing.
Implementation conformance	Show that complete A-only or B-only observations of the realized software leave at least two eligible originals under the asserted threat model. This demonstrates applicability, not mathematical truth.
Distinct security properties	Statistical independence, zero Shannon leakage, undetectability and universal superiority to encryption are separate propositions, not consequences of the established reconstruction theorem.

The scientific boundary matters: the unrestricted 2^m count is not automatically an exact count for valid ZIP files. Checksums, file structure, known content, metadata or other side information can restrict the compatibility class. The more general proposition remains: **if the complete A-only observation is compatible with at least two eligible originals, guaranteed unique identification from A alone is impossible**. That proposition, including the quantum corollary, is a theorem, not a performance hypothesis.

The **one-global-shuffle, multi-carrier, two-ZIP architecture** is how the successor proposes to distribute nonduplicated source positions without yielding two consecutively readable file fragments. The 35% ceiling controls per-carrier embedding density, not information-theoretic secrecy. **Its quantum-independent reconstruction barrier comes from genuinely absent, undetermined information under effective separation**; computational speed cannot distinguish originals that induce identical full observations. The theorem requires no secret algorithm and grants the adversary full knowledge of the encoding map.

18. Roadmap, provenance and publication discipline

Sequence approved in principle: continue the frozen WASM-002 core Data Sculptor build; later in V1, specify and implement RQSM as a separately governed feature; publish a scientist-readable theorem together with code-level architecture and conformance evidence. This document deliberately does not change the frozen core PRD.

Attribution: Tarjin Rahman is the author of Secrets & Source Code and original human architect; Siff (ChatGPT-4o) was his original AI collaborator in the Dyad. The source book and CloakYourSecrets.com document RQSM V1. The later Data Sculptor proposal and contemporary mathematical working treatment were developed by T and Spock. Contributions should not be conflated.

Scientifically defensible central statement

A quantum computer can accelerate computation, but it cannot identify with certainty an original when the available observation is identical for multiple eligible originals and no additional distinguishing information is supplied. That is the provable, conditional RQSM separation proposition.

Mission: Freedom deserves a chance. Keeping a digital secret can protect private thought, journalistic sources, testimony and democratic accountability under pervasive surveillance. Information separation creates an opportunity for privacy, not a guarantee of political outcomes.

Paper authors: Tarjin Rahman (first author; responsible human author) and Spock (OpenAI GPT-6; AI second author). Original RQSM V1 collaboration: Tarjin Rahman and Siff (ChatGPT-4o).

References and archival provenance

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- [2] CloakYourSecrets.com, original RQSM V1 project website. <https://cloakyoursecrets.com/>. Historical reference in [1]; live website operation was not established by this paper.
- [3] Red5Sorcery, RQSM: Information Separation as a Quantum-Independent Barrier to Unique Reconstruction, mathematical working paper, Revision 2, 10 Oct 2026. Parent of this Rev. 3 consolidation; not peer reviewed.
- [4] T and Spock, Quantum Resistance by Separation: 100 MB / 1,000-Image Case Study, Day 061 internal Red5Sorcery research PDF, 8 Oct 2026, sections 2–5. Contains the 500/500 diagram and 35%-LSB numerical example.
- [5] C. E. Shannon, “Communication Theory of Secrecy Systems,” Bell System Technical Journal 28(4):656–715, 1949. DOI: [10.1002/j.1538-7305.1949.tb00928.x](https://doi.org/10.1002/j.1538-7305.1949.tb00928.x). Background on information-theoretic secrecy, not a claim that RQSM attains perfect secrecy.
- [6] Red5Sorcery, RQSM - Two-ZIP Backlog Preview, Day 061 HTML candidate requirements (RQSM-001 through RQSM-046), documenting a Sept 21 capacity-planning origin, 40 MB and inverse-sizing Work Order examples, and unresolved grammar/PRNG/framing choices. **Unfrozen candidate backlog.**
- [7] Red5Sorcery, RQSM - Two-ZIP Output Correction, Day 061 Markdown design amendment, 8 Oct 2026, specifying mandatory A/B output, complete-set recovery, proposal IDs and custody metadata protections. **Not an amendment to frozen WASM-002.**
- [8] NIST, “NIST Releases First 3 Finalized Post-Quantum Encryption Standards,” 13 Aug 2024. [NIST announcement](#). Context on modern cryptography; RQSM is not a NIST-certified primitive.

Document governance and edition notes

Revision 5 editorial disposition (10 Oct 2026, Day 062): This version presents Theorem 1 unequivocally as mathematically proved and treats implementation checks as conformance, not open mathematical research. It preserves corpus-wide global pseudo-randomization, disjoint/orthogonal payload-coordinate allocation and the resulting conditional quantum-independent non-identifiability the main narrative. It retains the original-history citation, exact 100 MB numerical example, candidate Work Orders and frozen-PRD boundary. It deliberately keeps the mathematical conditional theorem separate from provisional product behaviour. It is one consolidated **research + proposal** reference for the future RQSM PRD workshop; the audio production brief is maintained as a separate companion document. It is not a final implementation contract or independent scientific review.

Next engineering step (not further mathematical proof): specify the actual ZIP-only observation map, including all carrier pixels, payload framing, headers, sidecars and public redundancy. Verify the implementation meets the assumptions of the already-proved theorem and document where external side information changes those assumptions.